

Automation in Penetration Testing

CompTIA PenTest+ (PT0-003)

Domains:

- 2.0: Reconnaissance and Enumeration
- 4.0: Attacks and Exploits

Objectives:

- 2.3: Given a scenario, modify scripts for reconnaissance and enumeration.
- 4.10: Given a scenario, use scripting to automate attacks.

Video Overview

This video will introduce you to the strategic role of automation in penetration testing, showing how it streamlines repetitive tasks and reduces human error. As you watch, consider how the principles of automating network scans, vulnerability checks, and even attack simulations can be applied to real-world cybersecurity challenges. The insights shared here will be directly applicable to your work in the field, helping you to enhance efficiency and precision in your future penetration testing engagements.

Key Topics Covered

- **Why Automate?** Streamlining repetitive tasks, reducing human error, and increasing efficiency
- **Getting Started with Scripts:** Automating ping sweeps with Bash scripts
- **Automating Scans:** Integrating and automating port and version detection scans with Nmap
- **Integrated Scripts:** Combining multiple tools like Nmap, Nikto, and GoBuster into single Bash scripts
- **Advanced Automation:** Developing custom Nmap Scripting Engine (NSE) scripts for specialized vulnerability enumeration
- **Automated Attack Tools:** Utilizing tools like PowerSploit, PowerView, PowerUpSQL, Impacket, and Scapy for PowerShell and Python-based attacks
- **Red Team Automation Tools:** Simulating adversary emulation with tools such as Caldera, Infection Monkey, and Atomic Red Team
- **Community Contributions:** Leveraging and contributing to open-source scripts on platforms like GitHub

Student Notes Section